

Here, $\varepsilon_{id} = Y_{0i} - m_d(p_i)$. The constant terms C_1 , C_2 and C_3 are those previously defined. The test statistic for hypothesis $H_{(A-4)}$ is

$$(\hat{m}_1(p) - \hat{m}_0(p))'(\hat{V}_0 + \hat{V}_1)^{-1}(\hat{m}_1(p) - \hat{m}_0(p)) \sim \chi_1^2.$$

Estimators for $\hat{V}_1$ and $\hat{V}_0$ are given below.

To test conditional independence and conditional mean independence of the residuals ($H_{(A-3)}$ and $H_{(A-4)}$ respectively), the test statistics are analogous except that Y_{0i} are replaced by estimated residuals $Y_{0i} - X_i\hat{\beta} = \hat{U}_i$. Tests of hypotheses $H_{(A-5)}$ and $H_{(A-5')}$, which test the identifying assumptions of conditional difference-in-difference matching estimators, are discussed below in the section on testing with panel data.

Variance estimators

First, we discuss the variance estimator for $\hat{V}_d$ in the test for tests of the conditional mean independence of earnings. One could estimate each of the components of the variance expression separately and construct a plug-in estimator, but instead we use an estimator proposed in Heckman *et al.* (1996b) that is more accurate at boundary points:

$$\hat{V}_d(p) = \sum_{i \in I_d} \tilde{\varepsilon}_{id}^2 \hat{W}_{id}^2(p),$$

where $\tilde{\varepsilon}_{id} = Y_{0i} - \hat{m}_d(p_i)$ and $\hat{W}_{id}(p)$ are weights equal to

$$\hat{W}_{id}(p) = \frac{G\left(\frac{P_i - p}{a_{N_d}}\right) \sum_{k \in I_d} G\left(\frac{P_k - p}{a_{N_d}}\right) (P_k - p)^2 - G\left(\frac{P_i - p}{a_{N_d}}\right) (P_i - p) \sum_{k \in I_d} G\left(\frac{P_k - p}{a_{N_d}}\right) (P_k - p)}{\sum_{i \in I_d} G\left(\frac{P_i - p}{a_{N_d}}\right) \sum_{k \in I_d} G\left(\frac{P_k - p}{a_{N_d}}\right) (P_k - p)^2 - \left[\sum_{k \in I_d} G\left(\frac{P_k - p}{a_{N_d}}\right) (P_k - p) \right]^2}.$$

The weights $\hat{W}_{id}(p)$ are the same as the local linear regression weights given in the text by equation (11), except that X is replaced by p . For the test of conditional independence at a point y_0 conditional on $P = p$, the estimators for the variance are analogous except that $\tilde{\varepsilon}_{id}$ is replaced by $\tilde{\varepsilon}_{id} = 1(Y_{0i} < y_0) - \hat{F}_d(p_i)$.

For tests of conditional independence of residuals, $\tilde{\varepsilon}_{id}$ is replaced by $\hat{U}_{0i} - \hat{E}(U_{0i} | P = p, D = d)$. For tests of conditional mean independence of residuals $\hat{\varepsilon}_{id} = 1(\hat{U}_{0i} < u_0) - \hat{E}(1(\hat{U}_{0i} < u_0) | P = p, D = d)$. Each of the conditional means is estimated by local linear regression.

Generalization of the test statistics to panel data and tests for conditional difference-in-difference matching ($H_{(A-5)}$, $H_{(A-5')}$)

In our empirical work, we use panel data and we conduct tests of conditional independence and conditional mean independence jointly across time periods and across p -points. Let i denote the individual and $t \in \{1, \dots, T\}$ the time periods in the panel. The generalization of the test for conditional mean independence of earnings ($H_{(A-4)}$) to one that is joint across time periods is:

$$[\tilde{m}(p)]'(\hat{V}_0^T + \hat{V}_1^T)^{-1}[\tilde{m}(p)] \sim \chi_T^2,$$

where $\tilde{m}(p) = \{\{\hat{m}_{11}(p) - \hat{m}_{01}(p)\}, \dots, \{\hat{m}_{1T}(p) - \hat{m}_{0T}(p)\}\}'$, with the first subscript denoting the group and the second subscript the time period. $\hat{V}_d^T$ is the T by T estimators of the variance-covariance matrix for group d . (Estimators of the covariance matrices are proposed below.) If the p -points are at least one bandwidth apart, the chi-squared statistics can be combined across p -points for an overall joint test across time periods and across quarters.

We next discuss tests of the identifying assumptions of conditional difference-in-difference methods (hypotheses $H_{(A-5)}$ and $H_{(A-5')}$). Let L denote the restriction matrix that takes differences $\hat{m}_{1t'}(p) - \hat{m}_{0t'}(p) - [\hat{m}_{1t}(p) - \hat{m}_{0t}(p)]$, where t' and t refer to a time period before and after the time of random assignment/eligibility determination, respectively, which are chosen to be symmetric around $t = 0$. For example, to test $H_{(A-5)}$ over time periods $t \in \{4, 5, 6\}$, define

$$L = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \end{bmatrix},$$

and the test statistic is

$$(L \cdot (\tilde{m}(p)))'(L \cdot (\hat{V}_0^T + \hat{V}_1^T) \cdot L')^{-1}(L \cdot \tilde{m}(p)) \sim \chi_k^2,$$

where k is the number of restrictions imposed by L . For testing hypothesis $H_{(A.5')}$, the test statistic is analogous except that Y_i are replaced by estimated residuals $Y_i - X_i\hat{\beta} = \hat{U}_i$.

Variance estimators

For the test of conditional mean independence jointly over all time periods at a point $P=p$, a natural estimator of the variance-covariance matrix is

$$\hat{V}_d^*(p) = \sum_{i \in I_d} \bar{\varepsilon}_i \bar{\varepsilon}_i' \hat{W}_{id}^2(p),$$

where $\bar{\varepsilon}_i = [\bar{\varepsilon}_{i1}, \dots, \bar{\varepsilon}_{iT}]$ and $\hat{W}_{id}(p)$ is the scalar weight defined above. The weights do not vary across time because P_i is constant across time for each individual. However, as discussed in Heckman, Ichimura, Smith and Todd (1996b), this estimator is infeasible when the panel is not balanced, as is the case in our data. If we construct consistent estimates of each component in the variance-covariance matrix, we cannot guarantee that the resulting estimated covariance matrix will be positive definite. The alternative estimator proposed in Heckman, Ichimura, Smith and Todd (1996b) that we use, which is always guaranteed to be positive semidefinite, is:

$$\hat{V}_d^T(p) = \sum_{i \in I_d} \hat{V}_{id} \hat{V}_{id}',$$

where

$$\hat{V}_{id} = [\bar{\varepsilon}_{i1}q(i, 1)\hat{W}_{id}(p), \dots, \bar{\varepsilon}_{iT}q(i, T)\hat{W}_{id}(p)]',$$

and where $q(i, t) = 1$ if observation i has usable data in period t and $q(i, t) = 0$ otherwise.

B. Accounting for parameter estimation

Conditional mean independence tests based on estimated residuals

When tests for conditional mean independence are conducted on estimated residuals, there is an additional source of estimation error. The fitted residuals come from estimating the following model for outcomes in the no treatment state:

$$Y_{0it} = X_{it}\beta + D_i E(U_{0it}|X_{it}, D_i = 1) + (1 - D_i)E(U_{0it}|X_{it}, D_i = 0) + \{U_{0it} - E(U_{0it}|X_{it}, D_i)\},$$

where the term in braces is a mean zero disturbance term by construction. The variables bear an it subscript because of the panel nature of the data. β is assumed to be the same across time. The estimation method for this type of partially linear model, which has both a parametric and a nonparametric component, is described in Heckman, Ichimura, Smith and Todd (1996b, c). In estimating this model, we draw on research reported in Heckman, Ichimura, Smith and Todd (1996b) who establish that in our data index sufficiency characterizes the conditional mean function, i.e. that $E(U_{0it}|X_{it}, D_i)$ has an index representation $E(U_{0it}|P(Z_i), D_i)$.

Estimates of the residuals for nonparticipants ($D_i = 0$) and for randomized-out controls ($D_i = 1$) are $\hat{U}_{0it} = Y_{0it} - X_{it}\hat{\beta}$. For the test of conditional mean independence of the residuals ($H_{(A.4')}$ in the text), the estimation of $\hat{\beta}$ should not affect the asymptotic distribution of the test statistic, because $\hat{\beta}$ converges at a faster rate than $\hat{E}(U_{0it}|P=p, D=d)$ (i.e. at rate $\sqrt{N}$ vs. $\sqrt{Na_N}$).

Table B-1 presents evidence on the importance of adjusting for parameter estimation error in constructing test statistics for testing $H_{(A.4')}$. The p values in the adjacent pairs of columns corresponding to unadjusted and adjusted test statistics differ dramatically. Both are asymptotically equivalent. In our view the truth is somewhere in between. Adjusting for parameter estimation greatly increases the sampling variances and makes it easy to accept any null. The less-than-perfect performance of matching based on Assumption (A-4') that is documented in the text makes us wary of relying uncritically on p -values in selecting a model.

C. Operational definition of common support and a Monte Carlo study of the relationship between sample size, trimming sensitivity and bandwidth sensitivity

In implementing nonparametric matching estimators, it is necessary to define a region of overlap. Using the notation defined in the text, the region S_{10} over which the supports of P overlap for the $D=1$ and $D=0$ groups is the region where $f(P|D=1) > 0$ and $f(P|D=0) > 0$. To operationally determine S_{10} , we first estimate the densities at all the sample P values using a kernel density estimator and determine $\hat{S}_{10}$ by forming

$$\hat{S}_{10} = \{P \in \hat{S}_1 \cap \hat{S}_0 : \hat{f}(P|D=1) > 0 \text{ and } \hat{f}(P|D=0) > 0\},$$

TABLE B-1
*P-values from tests for conditional mean independence of residuals**
(Asymptotic standard errors not adjusted and adjusted for estimation of β)
 $(H_{(\Delta_{\Delta})}) : E(U_0 | D=1, P) = E(U_0 | D=0, P)$

<i>P</i> -points	Residuals not adjusted		Residuals adjusted		Residuals not adjusted		Residuals adjusted	
	Pre-programme quarters†	Post-programme quarters‡	Pre-programme quarters†	Post-programme quarters‡	Pre-programme quarters†	Post-programme quarters‡	Pre-programme quarters†	Post-programme quarters‡
Adult males								
0.0025	0.0293	0.0002	0.6421	0.0955	0.1472	0.0363	0.8947	0.4683
0.005	0.0815	0.0004	0.7346	0.1129	0.1847	0.0298	0.8850	0.3761
0.01	0.2586	0.0040	0.7773	0.2170	0.3035	0.0140	0.8441	0.1377
0.02	0.4078	0.2056	0.6738	0.5925	0.3836	0.0180	0.6531	0.0532
0.03	0.5680	0.6563	0.7808	0.8289	0.1048	0.1243	0.4789	0.4907
0.04	0.7177	0.9060	0.8726	0.9563	0.0790	0.2570	0.4857	0.9276
0.05	0.8064	0.9885	0.9042	0.9939	0.2718	0.4510	0.6226	0.9325
0.10	0.7456	0.2591	0.9253	0.5790	0.4752	0.8423	0.7448	0.8878
Overall	0.2515	0.0001	1.0000	1.0000	0.0375	0.0008	1.0000	1.0000
Female youth								
0.0025	0.3877	0.2142	0.8999	0.4387	0.1418	0.3047	0.2058	0.4833
0.005	0.3547	0.2911	0.9403	0.5956	0.0770	0.3648	0.1323	0.5715
0.01	0.3373	0.4894	0.9697	0.8440	0.0274	0.5512	0.0668	0.7702
0.02	0.4702	0.6166	0.9616	0.7575	0.0107	0.4803	0.0561	0.7852
0.03	0.6913	0.5214	0.9378	0.5572	0.0277	0.1570	0.1654	0.4659
0.04	0.7342	0.4567	0.8816	0.4970	0.2278	0.0575	0.5479	0.2882
0.05	0.5988	0.4315	0.7717	0.4340	0.6279	0.0245	0.8679	0.2014
0.10	0.0000	0.1944	0.0021	0.6329	0.2412	0.0000	0.6887	0.0024
Overall	0.0015	0.3561	1.0000	1.0000	0.0006	0.0001	1.0000	1.0000

* A fixed bandwidth of 0.06 and biweight kernel, defined in Appendix A, are used in the test. The models for the probability of participation, P , are those described in the footnote on Table 2.

† Tests are performed jointly across pre-programme quarters $t = -1$ to $t = -6$.

‡ Tests are performed jointly across post-programme quarters $t = 1$ to $t = 6$.

where $\hat{S}_1$ and $\hat{S}_0$ are the estimated smoothed supports.⁴² The formal theory presented in our companion paper requires that the densities be strictly greater than 0, and in practice matches made at points P where the comparison group density is extremely small are likely to be inaccurate. We therefore require that the points that fall within the smoothed support $\hat{S}_{10}$ have a positive density that exceeds zero by a certain amount, determined by a "trimming level" $\hat{q}$. The set of points potentially eligible for matches are

$$S_q = \{P \in \hat{S}_{10} : \hat{f}(P|D=1) > c_q \text{ and } \hat{f}(P|D=0) > c_q\},$$

where c_q satisfies

$$\sup_{c_q} \frac{1}{2J} \sum_{i \in \bar{I}_1} \{1(\hat{f}(P_i|D=1) < c_q) + 1(\hat{f}(P_i|D=0) < c_q)\} \leq q,$$

where $\bar{I}_1$ is the set of observed values of P that lie in $\hat{S}_{10}$ and J is the cardinality of $\bar{I}_1$. Actual matches are made at the control P points (i.e. points for which $D=1$) in the original unsmoothed control point set, $\hat{S}_1$, that lie in S_q , i.e. $\hat{S}_1 \cap S_q$.

In conducting our empirical analysis we have explored the sensitivity of our estimates to choices of q . Monte Carlo evidence demonstrates the importance of trimming when sample sizes are small as they are for male youth. Smaller samples require higher trimming levels to reduce bias. There is no sensitivity to wide ranges of trimming rules for estimates obtained for adult samples or for female youth. (The empirical results are available on request from the authors).

Acknowledgements. The work reported here is a distant outgrowth of numerous papers and conversations with Ricardo Barros, Bo Honoré, and Richard Robb. We thank Manuel Arellano, three referees and Ed Vytlačil for helpful comments. We thank Annie Zhang and Jingjing Hsee for careful programming assistance. An earlier version of this paper circulated under the title "Evaluating The Impact of Training on the Earnings and Labor Force Status of Young Women: Better Data Help A Lot" (with Rebecca Roselius, 1993) and was presented at the Review of Economic Studies conference on evaluation research in Madrid in September, 1993. This paper was also presented by Heckman in his Harris Lectures at Harvard, November, 1995, at the Latin American Econometric Society meeting in Caracas, Venezuela, August, 1994; at the Rand-UCLA workshop in September, 1994; at UC Riverside, September, 1994; at Princeton, October, 1994; and at Texas, Austin, March, 1996.

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42. We use Silverman's "rule of thumb" bandwidth in estimating the densities. See equation 3.31 in Silverman (1986).